



Inverses

The graphs of function inverses are reflections of each other over the line $y = x$.

Visualizing the Inverse of a Graph

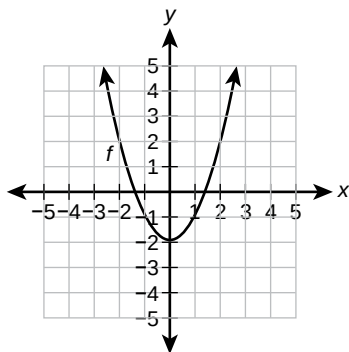
Relations and graphs have inverses.

Definition

The **inverse of a relation** switches the x - and y -values of the ordered pairs of the original relation. The domain of the inverse relation is the range of the original relation. The range of the inverse relation is the domain of the original relation. The graph of the inverse of a relation is its reflection over the line $y = x$.

EXAMPLE 1

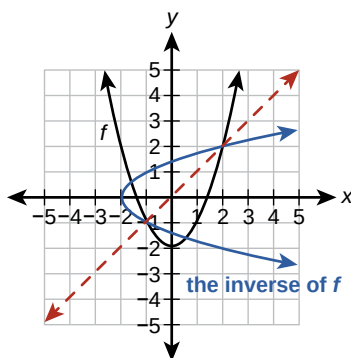
Graph the inverse of the function $f(x) = x^2 - 2$.



► **Remember** A function is a relation for which each input maps to a unique output.

SOLUTION

To create the graph of the inverse, reflect the graph over the line $y = x$. That will switch the x - and y -values of the ordered pairs. The points $(0, -2)$ and $(-2, 2)$ on the original graph become the points $(-2, 0)$ and $(2, -2)$ on the inverse of the graph.



Using the Horizontal Line Test

Some functions will have inverses that are functions while others will not.

Definitions

A function f is **invertible** if its inverse is also a function.

The **inverse of a function f** is denoted by f^{-1} .

To determine whether a function f is invertible, you could use the **horizontal line test**. If a horizontal line intersects a function more than once, then the function is not invertible—that is, the inverse of the function is not a function.

However, if every horizontal line intersects a function only once, then the function is invertible.

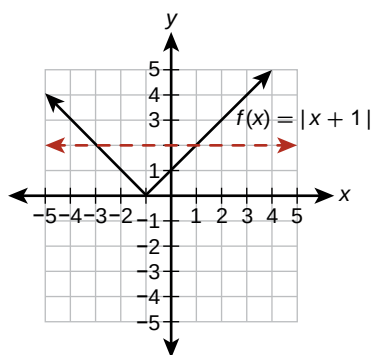
► **Remember** The vertical line test helps determine whether a relation is a function.

EXAMPLE 2

Determine whether $f(x) = |x + 1|$ is invertible.

SOLUTION

Graph the function.



There are many horizontal lines (such as the line $y = 2$ drawn on the graph) that intersect the graph at two points. The function is not invertible because it fails the horizontal line test. ■

Determining the Inverse of a Function

How to Determine the Inverse of a Function

Step 1 Replace $f(x)$ with y .

Step 2 Switch x and y .

Step 3 Solve the new equation for y .

Step 4 Replace y with $f^{-1}(x)$.

EXAMPLE 3

Determine the inverse of the function $f(x) = 2x - 1$.

SOLUTION

Step 1 Replace $f(x)$ with y .

$$y = 2x - 1$$

Step 2 Switch x and y .

$$x = 2y - 1$$

Step 3 Solve $x = 2y - 1$ for y .

$$y = \frac{x+1}{2}$$

Step 4 Replace y with $f^{-1}(x)$.

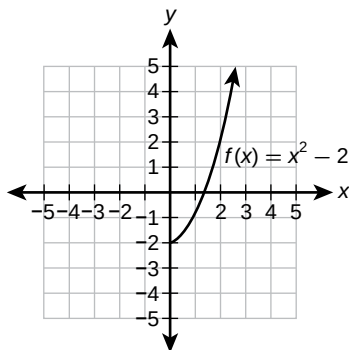
$$f^{-1}(x) = \frac{x+1}{2} \quad \blacksquare$$

Determining Functions with Restricted Domains

Some functions do not have inverses. However, they can have inverses when the domain is restricted.

EXAMPLE 4

Determine the inverse of $f(x) = x^2 - 2$, $x \geq 0$.



► **Remember** If a function is invertible, the domain of the function becomes the range of the inverse function. Similarly, the range of the function becomes the domain of the inverse function.

SOLUTION

The domain is $x \geq 0$, and the range is $y \geq -2$. The domain of the inverse is $x \geq -2$, and the range of the inverse is $y \geq 0$.

Determine the inverse.

$$y = x^2 - 2 \Rightarrow x = y^2 - 2 \quad \text{Switch } x \text{ and } y.$$

$$y = \pm\sqrt{x+2} \quad \text{Solve for } y.$$

Only $y = +\sqrt{x+2}$ has a domain of $x \geq -2$ and a range of $y \geq 0$.

Therefore, $f^{-1}(x) = \sqrt{x+2}$. ■